INCH-POUND

MIL-D-53092(ME)
7 February 1990

MILITARY SPECIFICATION

DRUM, FABRIC, COLLAPSIBLE, LIQUID FUEL, CYLINDRICAL, 500-GALLON CAPACITY, LOW TEMPERATURE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of nonvented cylindrical, collapsible elastomer drum, 500-gallon capacity, for storing and transporting fuels at all ambient temperatures between -60 and +95 °F.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel Oil, Diesel.
PPP-B-35	- Bags, Textile, Shipping, Burlap, Cotton, and Waterproof Laminated.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8110
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PPP-C-795 - Cushioning Material, Flexible Cellular, Plastic Film, for Packaging Applications.
PPP-P-291 - Paperboard, Wrapping and Cushioning.
PPP-T-60 - Tape: Packaging, Waterproof.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-G-3056 - Gasoline, Automotive, Combat.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-T-6396 - Tank, Aircraft Propulsion Fluid System, Internal, Removable, Non-Self-Sealing.
MIL-L-10547 - Liner, Case, and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible.
MIL-T-25959 - Tie Downs, Cargo, Aircraft.
MIL-G-46015 - Gasoline, Automotive, Combat, Referee Grade.
MIL-T-83133 - Turbine Fuel Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-889 - Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

TA13228E1860 - Drum, Fabric, Collapsible, Liquid Fuel, 500 Gallon Capacity, Low Temperature.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FS, Ft. Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the

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DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 381 - Existent Gum in Fuel by Jet Evaporation.
D 395 - Rubber Property - Compression Set.
D 412 - Rubber Properties in Tension.
D 471 - Rubber Property - Effects of Liquids.
D 573 - Rubber - Deterioration in an Air Oven.
D 751 - Coated Fabrics.
D 814 - Rubber Property - Vapor Transmission of Volatile Liquids.
D 910 - Aviation Gasolines.
D 1149 - Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimens).
D 2240 - Rubber Property - Durometer Hardness.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The drums shall be in accordance with TA13228E1860 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the drum. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and as shown on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.3.1 Material deterioration prevention and control. The drum shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which drum may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.1.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the drum may be newly fabricated from recovered materials to the maximum extent practicable, provided the drum produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the drum.

3.3.2 Drum body. The drum body shall be cord-reinforced low temperature elastomer construction with an inner liner and an outer cover. Liner and cover compounds shall meet the requirements of table I. Cord shall be multi-ply polyester. Carcass rubber shall be the same as, or compatible with, the liner and cover compounds. Use of solvents such as MEK to augment building tack of the carcass compound during drum fabrication is allowed. 1/ The color of the outer cover shall approximate Color No. 37030 of FED-STD-595. The drum shall be suitable for use with hydrocarbon fuels conforming to MIL-T-5624, MIL-G-3056, MIL-T-83133, VV-F-800, and ASTM D 910.

3.4 Drum structure and weight.

3.4.1 Puncture resistance. The drum wall (excluding the reinforced ends) shall withstand a force of not less than 200 pounds without being punctured when tested in accordance with 4.5.2.1.4.

3.4.2 Expansion dimensions. Drum dimensional changes shall be not more than the following, when subjected to a pressure of 30 pounds per square inch gage (psig) for 7 hours, when tested in accordance with 4.5.2.2.1.

1/ NOTE: Suggested materials and procedures for drum fabrication may be found in Report No. 5495410486, published by Goodyear Tire and Rubber Company, 1144 East Market Street, Akron, OH 44316, and Report No. ES-83-C125-1, published by Uniroyal, Incorporated, 312 North Hill Street, Mishawaka, IN 46544, and are available under Reports No. AD-B113563L and AD-A139329, respectively, from the Defense Technical Information Center, Cameron Station, Alexandria, VA 22304-6145.

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a. Overall length - 1 inch.
b. Diameter - 6 inches.

3.4.3 Weight. The weight of the completely assembled drum shall be not more than 275 pounds maximum when tested in accordance with 4.5.2.2.2.

TABLE I. Requirements for drum compounds.

Item	Property	Outer cover	Inner liner	ASTM method/ ref. paragraph
1	Tensile strength - psi (min)	1500	1500	D 412
2	Elongation - % (min)	250	250	D 412
3	Hardness - Shore A	60±5	60±5	D 2240
4	Heat resistance - 70 hours @ 100 °C			
	Hardness change - pts (max)	+15	+15	D 573
	Tensile strength change - % (max)	-20	-20	D 573
	Elongation change - % (max)	-40	-40	D 573
5	Fuel immersion - 70 hours @ 23 °C in ASTM ref fuel B			
	Volume change - % (max)	-	+50	D 471
6	Oil immersion - 70 hours @ 100 °C in ASTM oil No. 3			
	Volume change - % (max)	+120	+50	D 471
7	Ozone resistance - 7 days @ 40 °C 50 mPa and 20% elongation	No crack-ing	No crack-ing	D 1149
8	Compression set % (max) 22 hours @ 100 °C	80	50	D 395
9	Fuel contamination			
	Unwashed existent gum - mg (max)	-	60	D 381/4.5.2.1.2
	Heptane washed existent gum - mg (max)	-	7	D 381/4.5.2.1.3
10	Fuel permeability - fl oz/sq ft/ 24 hours (max)	-	1.5	D 814/4.5.2.1.1
11	Adhesion			
	Between cover and cord plies - lb/in. (min)	10	-	D 751
	Between cord plies - lb/in. (min)	10	-	D 751
	Between liner and cord plies - lb/in. (min)	-	10	D 751
12	Adhesion after immersion - 70 hours @ 23 °C in ASTM ref fuel B			
	Between liner and cord plies - lb/in. (min)	5	-	D 751
	Between cord plies - lb/in. (min)	5	-	D 751

3.5 Performance.

3.5.1 Ultimate hydrostatic pressure. The drum shall withstand an ultimate hydrostatic pressure of 45 psig, without any visual evidence of leakage, external or internal component layer separation, delamination, or blistering.

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3.5.2 Proof pressure. Each drum shall withstand a hydrostatic pressure of 30 psig without any visual evidence of leaking.

3.5.3 Fuel storage. The drum shall withstand the 72-hour fuel storage test without any visual evidence of leakage, inner liner separation, cracks, splits, or deterioration.

3.5.4 Air drop. The drum shall withstand three successive airdrops from a height of not less than 12 feet 6 inches without any visual evidence of leakage, broken cables or hardware, or separation of component layers of the drum body.

3.5.5 Towing, ground. The drum shall be capable of being towed as a wheel for 10 miles over paved road without any visual evidence of leakage.

3.5.6 Low temperature collapsibility. The drum shall be collapsible and capable of being emptied at -60 °F (-51 °C) with not more than 20 gallons of fuel remaining in the emptied drum. The drum shall show no visual evidence of leakage, blistering, delamination, splits, or cracks, chipping, or sloughing.

3.5.7 Ambient/chamber temperature collapsibility. The drum shall be filled and emptied through 80 cycles at all temperatures between -60 and +95 °F without any visual evidence of leakage, external or internal blistering, delamination, splits, cracks, or chipping.

3.5.8 Deceleration. The completely assembled drum shall withstand an average deceleration force of 8 times the force of gravity (g's) for a period of 0.1 seconds when tested as specified in 4.5.2.2.9 without any visual evidence of leakage, distortion, or fracture of hardware components.

3.5.9 Leakage. The drum shall withstand the leakage test specified in 4.5.2.2.10 without any visual evidence of leakage.

3.6 Marking for identification. Marking shall be as shown on drawing D13228E1848.

3.7 Workmanship. All metal parts shall be cleaned and free of sand, dirt, scale, and flux. Surfaces shall be smooth with edges rounded or beveled. The inside and outside of the drum shall be clean and free of foreign materials.

3.8 Valve labeling. The valve assembly shall be provided with a double-ended arrow showing the direction of operations and labeled at each end to indicate the functional result (i.e., OPEN and CLOSE).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.3 Disassembly inspection. Failure of any examination or test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.4 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.6 and 6.7).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).
d. Inspection comparison (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article drums shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.3.

4.3.2 Tests. The first article drums and material samples shall be subjected to the tests marked "X" in column 1 of table II. Every test specified in 4.5.2.2, except 4.5.2.2.4 and 4.5.2.2.10, shall be conducted on one of three drums so that approximately one-third of the tests shall be performed on each drum. Tests specified in 4.5.2.2.4 and 4.5.2.2.10 shall be conducted on each of three drums. All tests specified in 4.5.2.1 shall be performed on the material samples. The material samples shall be representative of materials used to fabricate the drum body. Failure of any test shall be cause for rejection or for performing the inspection specified in 4.1.3.

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TABLE II. Test schedule.

First article	Quality conformance: Individual	Quality conformance: Sample	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	-	X	Tensile strength - psi (min)	4.5.2.1	Table I
X	-	X	Elongation - % (min)	4.5.2.1	Table I
X	-	X	Hardness - Shore A	4.5.2.1	Table I
X	-	X	Heat resistance - 70 hrs @ 100 °C Hardness change - pts (max) Tensile strength change - % (max) Elongation change - % (max)	4.5.2.1	Table I
X	-	X	Ozone resistance - 7 days @ 50 mPa and 20 % elongation	4.5.2.1	Table I
X	-	X	Adhesion Between cover & cord plies - lb/in. (min) Between cord plies - lb/in. (min) Between liner & cord plies - lb/in. (min)	4.5.2.1	Table I
X	-	X	Compression set - % (max) 22 hours at 100 °C	4.5.2.1	Table I
X	-	X	Permeability.	4.5.2.1.1	Table I
X	-	X	Unwashed gum.	4.5.2.1.2	Table I
X	-	X	Existent gum.	4.5.2.1.3	Table I
X	-	X	Puncture resistance.	4.5.2.1.4	3.4.1
X	-	X	Expansion test.	4.5.2.2.1	3.4.2
X	-	X	Weight.	4.5.2.2.2	3.4.3
X	-	X	Ultimate pressure.	4.5.2.2.3	3.5.1
X	X	X	Proof pressure.	4.5.2.2.4	3.5.2
X	-	X	Fuel storage.	4.5.2.2.5.1	3.5.3
X	-	X	Airdrop.	4.5.2.2.5.2	3.5.4
X	-	-	Rolling tow.	4.5.2.2.6	3.5.5
X	-	-	Low temperature collapsibility.	4.5.2.2.7	3.5.6
X	-	-	Ambient temperature collapsibility.	4.5.2.2.8	3.5.7
X	-	-	Deceleration.	4.5.2.2.9	3.5.8
X	X	X	Leakage.	4.5.2.2.10	3.5.9

4.4 Quality conformance inspection.

4.4.1 Lot. A lot shall consist of not more than 50 drums or one month's production, each drum having successfully passed the tests specified in 4.4.4.1.

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4.4.2 Sampling for examination and tests.

4.4.2.1 Drums. Three drums shall be selected at random from each lot for tests.

4.4.2.2 Drum body. A sample of drum body material of the approximate thickness of the drum body shall be prepared from the elastomer compounds and cord or fabric used in fabricating the drums of the lot. The drum body sample shall be cured equivalent to the drums.

4.4.2.3 Drum material. A sample of the drum liner material, approximately 0.080 inch thick, cured equivalent to the drum shall be prepared.

4.4.3 Examination. All three drums selected (see 4.4.2.1) shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.4 Tests.

4.4.4.1 Individual. Each drum shall be subjected to the tests marked "X" in column 2 of table II. Failure of either test shall be cause for rejection of the drum.

4.4.4.2 Samples.

4.4.4.2.1 Drums. Drums selected in accordance with 4.4.2 shall be subjected to the tests marked "X" in column 3 of table II as follows: One of the drums shall be tested as specified in 4.5.2.2.2 and 4.5.2.2.3. The remaining two drums shall be tested as specified in 4.5.2.2.5 and 4.5.2.2.1. Failure of any test shall be cause for rejection of the represented lot of drums.

4.4.4.2.2 Drum material samples. Samples selected in accordance with 4.4.2 shall be tested as specified in 4.5.2.1.1 through 4.5.2.1.4. Failure of any test shall be cause for rejection of the represented lot of drums.

4.5 Inspection procedure.

4.5.1 Examination. The drum shall be examined as specified herein for the following defects:

101. Any part (or component) not in accordance with the QAP requirements as shown on the drawings.
102. Dimensions not as specified.
103. Materials not as specified.
104. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments.
105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
106. Contractor does not have documentation available for identification of materials, material finishes, or treatment.
107. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the drum.
108. Inner liner elastomer compound not as specified.

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109. Outer cover elastomer compound not as specified.
110. Cord or fabric elastomer compound not as specified.
111. Outer cover color not as specified.
112. Visual evidence of delamination between coating plies or between plies.
113. Identification label and marking not as specified.
114. Workmanship not as specified.
115. Valve labeling not as specified.

4.5.2 Tests.

4.5.2.1 Drum material samples. All materials shall be tested in accordance with the ASTM methods listed in table I except for the following tests.

4.5.2.1.1 Permeability test. The permeability test shall be conducted in accordance with ASTM D 814, except that the sample shall be identical in construction to the drum body prepared in accordance with 4.4.2.2. This test may be performed using an aluminum cup with a suitable clamping device in lieu of the permeability jar specified in ASTM D 814. The test liquid shall be ASTM D 471, reference fuel B. If aluminum cups are used, it is suggested that a nylon solution (Elvamide 8061) be used to seal the edges. The average of three determinations shall be reported. Nonconformance to table I shall constitute failure of this test.

4.5.2.1.2 Unwashed gum. Cut a 5-gram specimen from the drum interior coating compound into 0.0625-inch squares and place in a flask containing 250 mL of ASTM D 471 reference fuel B and allow to stand for 48 hours at 73 ±3 °F. Decant and filter the contaminated fluid through Whatman 41H filter paper, or equal. Determine the unwashed gum in accordance with section 10 of ASTM D 381 using the air jet vaporizing medium and an evaporation time of 45 minutes. The average of three determinations shall be reported. Nonconformance to table I shall constitute failure of this test.

4.5.2.1.3 Existent gum. Samples from 4.5.2.1.2 may be used to determine the heptane-washed existent gum in accordance with section 10 of ASTM D 381. The average of three determinations shall be reported. Nonconformance to table I shall constitute failure of this test.

4.5.2.1.4 Puncture resistance test. The drum wall (excluding the reinforced ends) shall be puncture tested as specified in MIL-T-6396, except that the minimum force required shall be 200 pounds. Nonconformance to 3.4.1 shall constitute failure of this test.

4.5.2.2 Drums.

4.5.2.2.1 Expansion test. The drum shall be filled with water to a pressure of 30 psig. The length and diameter shall be measured within 15 minutes. The length shall be measured by placing a 90-degree square upright at each end and measuring the distance between squares. The diameter shall be similarly measured by placing the square upright against each side at the center of the drum and measuring the distance between squares. These measurements shall be the original dimensions. The drum shall be maintained at 30 psig for 7 hours; at the end of the 7 hours, the measurements of length and diameter shall be repeated as described above. The difference between these final measurements and the

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original measurements shall be regarded as the changes in dimensions. Nonconformance to 3.4.2 shall constitute failure of this test.

4.5.2.2.2 Weight test. The emptied drum with hardware shall be weighed. Nonconformance to 3.4.3 shall constitute failure of this test.

4.5.2.2.3 Ultimate pressure test. The drum shall be subjected to a hydrostatic pressure of 45 psig and allowed to stand for 30 minutes to determine compliance with 3.5.1. At the end of 5 minutes, the pressure shall be readjusted to 45 psig if the pressure has dropped due to expansion of the drum. After the hydrostatic test, the drum shall be emptied and visually examined, both externally and internally. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.2.4 Proof pressure test. Each drum shall be subjected to a hydrostatic pressure of 30 psig and allowed to stand for 30 minutes. At the end of 5 minutes, the pressure shall be readjusted to 30 psig if the pressure has dropped due to expansion of the drum. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.2.5 Fuel storage and airdrop test.

4.5.2.2.5.1 Fuel storage test. The drum shall be filled with a minimum of 450 gallons of gasoline conforming to MIL-G-3056, type II, and allowed to stand for a period of 72 hours at ambient temperature (40 to 90 °F). At the end of each 24-hour period, the drum shall be rotated 180 degrees and then examined for leakage. After completion of the 72-hour period. the drum shall be emptied and the hardware shall be removed for visual examination of the inner liner. Nonconformance to 3.5.3 shall constitute failure of this test. The gasoline shall be analyzed for existent and unwashed gum levels both before and after the storage and the results reported.

4.5.2.2.5.2 Airdrop test. After completion of the fuel storage test, the drum shall be reassembled and refilled with a minimum of 450 gallons of fluid having a specific gravity of between 0.72 and 1.00. The drum shall then be lifted by the two anchor shackles on the fill end until the opposite end has a minimum ground clearance of 12 feet 6 inches. The drum shall then free-fall onto unprepared ground such as grass, sand, or bare earth. The soil shall not contain excessive water to form a viscous liquid. After three free-falls, the drum shall be visually examined for leaks; then emptied and visually examined, internally and externally, for evidence of broken hardware and separation of component layers of the drum body. The time between draining the fluid and completion of the free-fall tests shall not exceed 3 hours. Nonconformance to 3.5.4 shall constitute failure of these tests.

4.5.2.2.6 Rolling tow test. The drum shall be filled with a minimum of 350 gallons of water. A suitable towbar shall be attached to the lugs, and the drum shall be towed as a wheel for not less than 10 miles over a paved road. The rate of speed of towing shall be between 5 to 10 miles per hour (mph). Nonconformance to 3.5.5 shall constitute failure of this test.

4.5.2.2.7 Low temperature collapsibility test. The emptied drum shall be cooled to -60 °F (-51 °C) and filled with a measured 460 gallons of gasoline conforming to MIL-G-46015, grade I, which has been cooled to -60 °F. The drum

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shall be cold soaked for 24 hours at -60 °F. The drum shall then be emptied of not less than 440 gallons of gasoline using a suitable pump and subjected to a minimum vacuum of 12 inches of mercury (Hg). This test shall be conducted at -60 °F with the drum in a horizontal position. Nonconformance to 3.5.6 shall constitute failure of this test.

4.5.2.2.8 Ambient/chamber temperature collapsibility test. The emptied drum shall be filled with a measured 460 gallons of gasoline conforming to MIL-G-46015, grade I kerosene, JP-5, JP-8 or diesel fuel. The drum shall then be emptied of not less than 435 gallons of gasoline, kerosene, JP-5, JP-8 or diesel fuel using a suitable pump and subjecting the drum to a minimum vacuum of 12 inches Hg. This test shall be conducted at the ambient chamber temperature with the drum in a horizontal position. The drum shall be filled and emptied through 80 cycles, with each filling and emptying constituting one cycle. The 80 cycles shall include 10 cycles at each range of temperatures, -60 to -50 °F, -49 to -40 °F, -39 to -30 °F, -29 to 0 °F, +1 to +20 °F, +21 to +45 °F, +46 to +70 °F, and +70 to +95 °F. The drum hardware shall be disassembled for examination of the drum internal carcass. Nonconformance to 3.5.7 shall constitute failure of this test.

4.5.2.2.9 Deceleration test. The drum shall be filled with liquid having a specific gravity between 0.72 and 1.00 to a hydrostatic pressure of 4.5 ±0.5 psig. The drum shall be mounted on a simulated aircraft cargo floor including standard 20-inch grid tiedown shackles. The drum shall be restrained by a new set of 10,000-pound pull chains fastened between drum shackles and simulated cargo floor. Two 10,000-pound pull chains conforming to MIL-T-25959 shall be secured to one drum shackle and the other ends of the chains shall be secured to two tiedown rings toward the rear of the simulated cargo floor at a distance of 58 ±1 inches from the longitudinal axis of the drum; the two tiedown rings shall be 20 ±1 inches apart. One 10,000-pound pull chain shall be secured to the other drum shackle and the other end of the chain secured to a forward tiedown ring on the simulated cargo floor at a distance of 35 ±1 inches from the longitudinal axis of the drum. These chains as specified shall be connected to the drum shackle on both ends of the drum. This test assembly shall be subjected to an average 8g deceleration force for a period of not less than 0.1 second. The drum shall be examined for leakage and the drum hardware shall be removed for visual examination. Water may be used in this test. When water is utilized, a reduction of "g" force consistent with the increased weight will be permitted (5.7g). Nonconformance to 3.5.8 shall constitute failure of this test.

4.5.2.2.10 Leakage test. Each drum shall be inflated with air to approximately 6 psig and sprayed with soap or detergent sudsing solution over the entire surface to determine compliance with 3.5.9. Leakage shall be determined by observing areas for fizzing or bubbling of soap or detergent sudsing. If any evidence of fizzing or bubbling is observed, a Freon test shall be made on the drum. Inflate drum to 2 psig with air, add Freon to bring pressure of 4 psig, and add additional air to bring the drum to to 6 psig and allow to stand for 15 minutes. At 5 minute intervals for 15 minutes, use a Halogen leak detector to determine if the drum is defective . Evidence of leakage determined by the Freon test shall constitute failure of this test.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of 1 or more defects shall be cause for rejection.

116. Drums not thoroughly drained and dried as specified for level A.
117. Shackles not wrapped and wrap not secured as specified for level A.
118. Drum not collapsed and folded as specified for level A.
119. Technical manual not preserved as specified.
120. Coupler valves not wrapped and wrap not secured as specified for level A.
121. Coupler valve and technical manual not placed in the bag specified and the bag not secured as specified for level A.
122. Shipping container not as specified for levels A, B, and C.
123. Box closure and strapping not as specified for levels A, B, and C.
124. Marking missing, illegible, incorrect, or incomplete for levels A, B, and C.

4.7 Inspection comparison. The Government may select drums at any time during the contract production period and subject these drums to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article drum. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of any drum to meet any requirement specified herein, during, and as a result of the examination and tests specified in 4.7, shall be cause for rejection of the drum and shall be cause for refusal by the Government to continue acceptance of production drums until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on drums produced and accepted under the contract. Any deficiencies found as a result of comparison inspection will be considered prima facie evidence that all drums accepted prior to completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.8). The first article pack may be accomplished utilizing either the first produced drum or a production drum. If the first produced drum is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first produced drum and the production drum.

5.2 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Drum. Each drum shall be thoroughly drained, the interior and exterior shall be dried and dusted with talc prior to packaging. Each shackle shall be wrapped with corrugated, single-faced paperboard material conforming to PPP-P-291, type I, style 1, which shall be secured with tape conforming to PPP-T-60, type IV. Each drum shall be collapsed and compactly folded so that one swivel plate is on the interior of the bundle and the other is on the top exterior of the bundle.

5.2.1.2 Components. The technical manual shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3. Each coupler valve shall be wrapped with cushioning material conforming to PPP-C-795, class I, having a thickness classified as thick. Cushioning material shall be secured with tape conforming to PPP-T-60, type IV. The technical manual and the cushioned coupler valves shall be placed in a bag conforming to PPP-B-35, type I or II, size as appropriate. The bag shall be secured with tie cord or wire to a lug on the exposed end swivel plate of the drum.

5.2.2 Level C.

5.2.2.1 Drums. Each drum shall be thoroughly drained and dried, then dusted with talc. The drum shall then be collapsed and folded as specified for level A (see 5.2.1.1).

5.2.2.2 Components. The components shall be preserved as specified for level A (see 5.2.1.2).

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.3.1 Level A. Each complete drum, preserved as specified in 5.2, shall be individually packed in a close-fitting box conforming to PPP-B-621, class 2, style 2 or 2-1/2, type 3 load. Each box shall be equipped with a case liner conforming to MIL-L-10547, type IV, grades E or F. When skids and reinforcement battens are required, the battens shall be placed on the box exterior and the skids shall be placed on the outside of the box bottom, in alignment with each side reinforcement batten. Box assembly, closure, and strapping shall be as specified in the appendix to the box specification except that the strapping shall be flat, the finish B, and a dado shall be incorporated into the bottom

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face of each skid, full length at the centerline, so that strapping, normally positioned adjacent to the skids, shall completely encompass the reinforcement battens and the skids. Alternatively, when specified (see 6.2), the shipping container may be a close-fitting box conforming to PPP-B-601, overseas type, style A or B, with case liner as specified herein and with the box lid secured to the box side and end panels with strapping only in lieu of nailing and strapping.

5.3.2 Level B. Each complete drum, preserved as specified in 5.2, shall be packed as specified in 5.3.1 for level A except that boxes shall be domestic type, or each complete drum, preserved as specified in 5.2.1, shall be packed in a close-fitting box conforming to PPP-B-640, class 2, style A, B, C, or D. Strapping for level B packing shall be finish A.

5.3.3 Level C. Each complete drum, with accessory components, shall be packed in a snug-fitting container appropriate for the weight and size of the drums. The container shall be capable of being handled by mechanical means and of being stacked during storage. Container closure shall allow reuse of the container.

5.4 Marking. Marking for levels A, B, and C shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 500-gallon collapsible drum is intended for use at all ambient temperatures between -60 and +95 °F for the following:

a. Storage, receipt and issue of liquid fuel.
b. Transporting liquid fuel by:

(1) Truck.
(2) Aircraft.
(3) Helicopter.
(4) Airdrop from fixed wing aircraft.
(5) Towing as a wheel.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of first article and number of drums required (see 3.2).
d. When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. Level of preservation and packing required (see 5.2 and 5.3).

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f. When alternate PPP-B-601, style A or B box with special closure is required for level A (see 5.3.1).

6.3 First article. When a first article inspection is required, the item(s) should be preproduction models. The first article should consist of three units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including and special tools) and instructions regarding shipment of drums.

6.6 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.7 Definition.

6.7.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP's indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.8 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the drums in accordance with this specification.

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6.9 Subject term (key word) listing.

Collapsible elastomer drum
Cord-reinforced
Drum
Drum body
Elastomer
Fuel
Inner liner
Low temperature environment
Outer cover

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 8110-A290

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-D-53092(ME)	900207

3. DOCUMENT TITLE: Drum, Fabric, Collapsible, Liquid Fuel, Cylindrical, 500-Gallon Capacity, Low Temperature

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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